Dynamic Welfare-Maximizing Group Counting with Count-Valued Outcomes:

A Separation, Tractable Regimes, and Empirical Evidence

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Abstract

A welfare-maximizing testing planner uses a limited budget of pooled tests to certify healthy individuals. Prior work studies static designs and dynamic policies with binary outcomes. We study a richer dynamic model in which a test reports the exact number of infected samples in its pool. The paper has three parts. First, we give a closed-form separation: in a homogeneous high-prevalence population, the best static design earns Buq, whereas a dynamic count-valued strategy searches kG people and earns at least $u[1-(1-q)^{kG}]$ under the same strict hard-clearing objective. Because the comparison changes both adaptivity and outcome resolution, we state it only as a joint separation. Second, we give efficient exact posterior algorithms for structured histories: componentwise inference for disconnected test hypergraphs, an elementary-symmetric-polynomial algorithm for fully observed laminar layers, and a linear-time forward–backward algorithm for an acyclic chain with one-bit separators. Third, exact experiments show mean relative improvements of 0.63%, 3.97%, and 5.07% on three small-instance regimes. A separate resolution experiment finds that the three-level channel $\{0, 1, \geq 2\}$ captures 84.5% of the full counting gain on a selected $N = 6$ instance. Together, these results identify when richer outcomes improve welfare, when exact inference remains tractable, and where the present analysis stops.

1 Introduction

Imagine a planner facing 32 people in a high-prevalence population. Each person is healthy with probability $q = 0.05$, a healthy certification is worth u, and the planner has seven tests. A fixed binary design does best by testing seven people individually, for expected welfare $7uq = 0.35u$. A count-valued policy can instead test two disjoint pools of 16. If either pool contains a healthy person, four count-guided splits isolate one such person, and a final singleton test certifies that person under the strict hard-clearing rule. This policy earns at least

$$u\big[1 - (1 - 0.05)^{32}\big] \approx 0.806u.$$

The mechanism is a counting argument. Individual tests inspect seven possible healthy people; count feedback turns the remaining tests into a search over 32.

This example motivates dynamic welfare-maximizing group counting. A planner has a heterogeneous population, a test budget, and a physical pool-size limit. Welfare is earned only for people placed in a pool observed to contain no infected samples. Static welfare-maximizing pooled testing fixes every pool in advance [1]. Dynamic binary testing chooses the next pool after observing whether each earlier pool was all-clear [2]. We ask what becomes possible when the observation is the full count of infected samples.

Count-valued observations are established in quantitative group testing, where the usual objective is recovery with as few tests as possible [4]. Semi-quantitative group testing similarly replaces a binary result with a small number of qPCR-derived bins [5]. Our novelty is not the count channel itself. It is the welfare objective under strict hard clearing, the separation from fixed designs, and the use of structured count-posterior inference inside a dynamic decision problem.

The example changes two ingredients at once: the policy becomes adaptive and the outcome becomes more informative. It therefore proves a joint separation, not a value-of-counting result in isolation. The unresolved benchmark is the optimal dynamic binary policy on the same homogeneous family; solving it would separate the gains from adaptivity and resolution.

We make three contributions.

1. We prove a closed-form joint separation under strict hard clearing, including the final certification test that the reward definition requires.
2. We give exact posterior algorithms for verified tractable structures: disconnected components, fully observed laminar layers, and an acyclic one-bit-separator chain. These algorithms expose the structural reason exact inference can scale without enumerating all 2^N latent profiles.
3. We consolidate two empirical views. A hierarchy experiment measures the dynamic count-versus-binary gap, while a resolution curve measures how much of that gap is captured by intermediate count channels.

We keep the main text focused on this separation–inference–evidence connection. Results specific to dynamic binary testing are inherited from the predecessor study [2]; posterior complexity, exact fiber enumeration, and unrestricted sampling on count fibers are discussed only in Appendix B or left for future work.

2 Model and Comparison Classes

There are N individuals. Individual i has utility $u_i \geq 0$ and an independent latent infection state $Z_i \sim \text{Bernoulli}(p_i)$. Write $q_i = 1 - p_i$ for the probability that i is healthy. A test may contain at most G individuals, and the planner may perform at most B tests.

For a pool $t \subseteq [N]$, the count-valued outcome is

$$r(t, Z) = \sum_{i \in t} Z_i.$$

The corresponding binary outcome is $\mathbf{1}[r(t, Z) > 0]$. The count channel refines the binary channel: the binary result can be recovered from the count, but not conversely.

A policy produces a history $h_B = ((t_1, r_1), \dots, (t_B, r_B))$. Under *strict hard clearing*, welfare is credited only after a person belongs to an actually tested zero-count pool:

$$U(h_B, Z) = \sum_{i=1}^{N} u_i \mathbf{1}[\exists j \leq B : \; i \in t_j, \; r_j = 0] \, .$$

Deduction alone is not rewarded. A policy that infers a person is healthy must still place that person in a zero-count test to collect the utility.

We compare three classes.

Definition 1 (Static binary design)**.** *A static binary design fixes all B pools before any outcome is observed. Because welfare depends only on zero versus nonzero outcomes, additional positive-count resolution cannot change the value of a fixed design.*

Definition 2 (Dynamic binary policy)**.** *A dynamic binary policy chooses t_j as a function of the previous pools and binary outcomes $\mathbf{1}[r_\ell > 0]$, $\ell < j$.*

Definition 3 (Dynamic count-valued policy)**.** *A dynamic count-valued policy chooses t_j as a function of the previous pools and exact outcomes r_ℓ, $\ell < j$.*

Let $\mathcal{U}^{\text{stat}}$, $\mathcal{U}^{\text{dyn,bin}}$, and $\mathcal{U}^{\text{dyn,count}}$ denote the optimal expected welfare in these classes. Since a dynamic policy can ignore its history and a count policy can coarsen its observations,

$$\mathcal{U}^{\text{stat}} \le \mathcal{U}^{\text{dyn,bin}} \le \mathcal{U}^{\text{dyn,count}}.$$

The hierarchy is definitional. The size of either gap is a substantive question.

3 A Separation for Dynamic Count-Valued Testing

We now formalize the example from the introduction. Consider a homogeneous population with common utility $u > 0$ and common healthy probability $0 < q < 1/2$. The population is otherwise exchangeable.

Lemma 1 (Static optimum)**.** *For the homogeneous instance above, if $N \ge B$, the optimal static design under strict hard clearing has value*

$$\mathcal{U}^{\text{stat}} = Buq.$$

This remains true when the fixed pools may overlap.

Proof. A pool of size g is all-clear with probability q^g and then credits gu, so its expected contribution is ugq^g. For $g \ge 1$ and $q < 1/2$,

$$gq^{g-1} \le \frac{g}{2^{g-1}} \le 1,$$

and hence $ugq^g \le uq$. Summing over B tests gives an upper bound Buq for disjoint pools.

For overlapping pools, write the welfare as a sum over individuals and use a union bound on the events that certify each individual. Reversing the order of summation again bounds the expected welfare contributed by each test by $ugq^g \le uq$. Thus overlap cannot exceed Buq. Testing B distinct individuals attains the bound. □

Theorem 1 (Joint static-binary versus dynamic-count separation)**.** *Let G be a power of two, let $0 < q < 1/2$, and let*

$$B = k + \log_2 G + 1$$

for an integer $k \ge 1$. If $N \ge \max\{B, kG\}$, then

$$\mathcal{U}^{\text{dyn,count}} \ge u\left[1 - (1-q)^{kG}\right].$$

Consequently, whenever

$$Bq < 1 - (1-q)^{kG},$$

the dynamic count-valued policy strictly outperforms every static design.

Proof. Use the first k tests on k disjoint pools of size G. If every person in those pools is infected, the construction earns nothing. Otherwise at least one tested pool has count strictly below its size and therefore contains a healthy person.

Maintain a block known to contain at least one healthy person. Split it into two equal halves and test one half. If its infected count is smaller than its size, keep that half. Otherwise it is entirely infected, so keep the other half. After $\log_2 G$ tests, the maintained block is a singleton known to be healthy. The last test places this singleton in an observed zero-count pool, which is necessary under strict hard clearing.

The policy certifies at least one healthy person exactly on the event that the initial kG people contain one. That event has probability $1-(1-q)^{kG}$. □

For $(q, G, k) = (0.05, 16, 2)$, the budget is $B = 7$. Lemma 1 gives $0.35u$, while Theorem 1 gives $0.806u$. The lower bound counts only one certified person, although zero-count pools encountered along the way may certify more.

For fixed B and power-of-two G, strict clearing gives $kG = (B - \log_2 G - 1)G$. Equivalently, $G = 2^{B-k-1}$, so integer $k \geq 1$ gives $kG = k2^{B-k-1}$. The maximum is 2^{B-2}, attained at $k = 1$ and also at $k = 2$ whenever that choice is feasible. The optimization sharpens the construction but is not needed for the separation.

4 Efficient Exact Inference in Structured Regimes

A dynamic policy must update beliefs after observing counts. Unrestricted histories couple many latent states, but simple overlap structure produces small separators. The algorithms below compute exact posterior marginals without enumerating every global profile. They are inference algorithms; integrating each representation into a globally optimal large-scale decision policy is a separate problem.

4.1 Disconnected components

Form a bipartite incidence graph between individuals and previously tested pools. If this graph has disconnected components, the posterior factorizes across them because the prior is independent and every count constraint lies inside one component.

Proposition 1 (Componentwise exact inference)**.** *If every tested connected component contains at most c individuals, exact posterior marginals can be computed in*

$$O\left(N + \sum_{C} 2^{|C|}(|C| + |h_C|)\right),$$

where h_C is the set of count constraints in component C. Thus the calculation is linear in the number of components for fixed c, rather than exponential in the full population size.

Proof. The likelihood indicator factors by component, as does the independent prior. Enumerate feasible profiles separately within each component and normalize locally. Products of the local normalizers recover the global normalizer, while a marginal depends only on its own component. □

This decomposition is effective when adaptive tests form many small disconnected regions. It provides exact inference within each region, but does not imply global optimality of a greedy pool-selection rule.

4.2 Fully observed laminar layers

Suppose the tested pools form a nested hierarchy and every relevant level has been observed. Subtracting child counts from parent counts fixes an exact count inside each disjoint layer. For example, if $B \subset A$, with counts two and five, then $A \setminus B$ contains exactly three infected individuals.

Consider one layer L of size m with observed infected count d. For $0 < p_i < 1$, define odds $w_i = p_i/(1-p_i)$ and let $e_d(w_L)$ be the elementary symmetric polynomial of degree d. When $d \geq 1$, conditioning on $\sum_{i \in L} Z_i = d$ gives

$$\Pr(Z_i = 1 \mid \textstyle\sum_{j \in L} Z_j = d) = \frac{w_i \, e_{d-1}(w_{L \setminus \{i\}})}{e_d(w_L)}.$$

When $d = 0$, every marginal in the layer is zero. Deterministic priors are removed and their contribution is subtracted from d before applying the formula. Prefix and suffix recursions compute all required symmetric polynomials and all m marginals in $O(m(d+1))$ time. Each laminar layer is then solved independently.

Proposition 2 (Fully observed laminar inference)**.** *When nested observations determine disjoint layers $L_1, \dots, L_s$ and their counts $d_1, \dots, d_s$, exact posterior marginals are computable in*

$$O\left(\sum_{\ell=1}^{s} |L_\ell|(d_\ell + 1) \right)$$

time and $O(\max_\ell |L_\ell|(d_\ell + 1))$ working memory.

The proposition applies only when every layer count is determined. Partially observed laminar trees require message passing over unresolved parent counts, which is outside the present algorithmic scope.

4.3 An acyclic one-bit-separator chain

Now consider the overlapping chain

$$\{0, 1, 2\}, \ \{2, 3, 4\}, \ \{4, 5, 6\}, \dots.$$

Adjacent factors share one person, so a message needs to remember only one binary state. A forward message $\alpha_j(s)$ stores the unnormalized mass of all assignments through pool j whose right boundary has state $s \in \{0, 1\}$. The transition sums over the one new interior state subject to the observed count. A backward pass yields every boundary and interior marginal.

Proposition 3 (Exact chain inference)**.** *For m size-three pools arranged as above, all posterior marginals can be computed exactly in $O(m)$ time and $O(m)$ memory.*

This is an acyclic factor graph with a one-bit separator. Its primal graph has treewidth two, so calling it simply a treewidth-one model would be ambiguous. More generally, junction-tree inference is polynomial in instance size at fixed separator width and exponential in that width and the relevant factor domain [6]. Our prototype implementation covers this chain special case; a general bounded-treewidth solver remains future work.

5 Empirical Evidence

The experiments answer two different questions. The hierarchy experiment compares optimal dynamic binary and count-valued policies. The resolution curve then asks how much outcome detail is needed to capture the count-valued gain.

Table 1: Exact dynamic binary and count-valued welfare. The $N = 3$ and $N = 5$ rows use 200 seeded instances; the heavier $N = 7$ row uses 40. Standard errors in the last column are in percentage points.

N	B	G	$\overline{\mathcal{U}}^{\text{dyn,bin}}$	$\overline{\mathcal{U}}^{\text{dyn,count}}$	relative gain $\pm$ SE
3	2	3	2.686	2.704	$0.63\% \pm 0.09$
5	3	5	4.600	4.771	$3.97\% \pm 0.19$
7	3	7	5.454	5.717	$5.07\% \pm 0.37$

Table 2: Resolution curve on the selected $(N, B, G) = (6, 3, 4)$ instance.

channel	optimal welfare	fraction of count gain
$\{0\}/\{\geq 1\}$	4.967923	0.000
$\{0\}/\{1\}/\{\geq 2\}$	5.297456	0.845
$\{0\}/\{1\}/\{2\}/\{\geq 3\}$	5.357733	1.000
full count	5.357733	1.000

5.1 Dynamic binary versus dynamic count

For each configuration, infection probabilities are drawn independently from $U(0, 1)$ and utilities from $\{1, 2, 3\}$. The exact solvers compute the complete value hierarchy. Table 1 reports the resulting sample means. The percentage is computed instance by instance and then averaged: $100(\mathcal{U}^{\text{dyn,count}} - \mathcal{U}^{\text{dyn,bin}})/\mathcal{U}^{\text{dyn,bin}}$. It is not the ratio of the two aggregate means. The generator is `augmented/hierarchy_experiment.py` with PRNG seed 42; raw rows are stored in `hierarchy_small.csv` and `hierarchy_n7.csv` under `results/hierarchy`.

The hierarchy inequalities hold on every instance. The count-valued optimum is strictly larger in 93 of 200, 196 of 200, and 40 of 40 instances, respectively; the median relative gain is zero in the smallest regime. Because N, B, and G vary together, these rows motivate rather than replace a controlled sweep over the three parameters.

5.2 Resolution as a mechanism experiment

A truncation channel reports $\min(r, c)$. The endpoint $c = 1$ is binary, while $c \geq G$ is the full count. The sweep contains eight designed instances. We report `mixta_n6_g4` because it is the only recorded curve with a nontrivial interior fraction, not as an estimate of how often such a curve occurs. Its parameters are $p = (0.15, 0.25, 0.35, 0.45, 0.55, 0.65)$, $u = (1, 2, 3, 1, 2, 3)$, $B = 3$, and $G = 4$. On this instance,

$$\text{frac}(c) = \frac{\mathcal{U}_{\text{c}} - \mathcal{U}_{\text{bin}}}{\mathcal{U}_{\text{count}} - \mathcal{U}_{\text{bin}}}.$$

The exact values appear in Table 2.

Thus the three-level channel captures 84.54% of the full binary-to-count difference on this instance. This is a mechanism result on a selected small case, not a population average. Other designed instances saturate at the same or an earlier truncation level.

5.3 Structured-inference checks

The notebook compares the prototype structured routines with exhaustive enumeration on small instances and then evaluates the same routines where global enumeration is impossible. Table 3

Table 3: Executed correctness and scale checks for structured exact inference.

structure	instance	maximum marginal error	time
laminar layers	$N = 12$ versus brute force	2.2×10^{-16}	< 1 ms
laminar layers	$N = 6000$	global enumeration infeasible	13 ms
one-bit chain	$N = 13$ versus brute force	1.1×10^{-16}	not recorded
one-bit chain	200 pools, $N = 401$	global enumeration infeasible	1.1 ms

records those single-instance checks, not a validation suite. Timings are single-run measurements on one machine and are only illustrative.

6 Discussion and Limitations

Two mechanisms remain coupled. The separation is between static binary and dynamic count-valued testing. It does not yet quantify the optimal dynamic-binary middle cell on the same homogeneous family. The empirical hierarchy includes that cell on finite heterogeneous instances, but it does not solve the analytic family in Theorem 1.

The algorithms address inference rather than global policy optimization. The component, laminar-layer, and chain routines compute exact posterior marginals in their stated regimes. These marginals can support greedy scoring or a small-state dynamic program, whereas a scalable exact policy optimizer must also control the future action space.

Exact counts are an idealization. A quantitative laboratory measurement is noisy and often reports a range or a continuous signal rather than an integer infection count. The resolution experiment gives a first bridge to coarser channels, but a calibrated observation model and external data are required before making a screening recommendation.

The experiments are small and synthetic. The exact comparison is valuable because it separates policies without approximation error. Its population sizes are nevertheless limited by exact dynamic optimization. The increasing percentages in Table 1 should be treated as evidence for a mechanism, not as an estimate for a particular deployment.

7 Conclusion

Count-valued outcomes can change the economics of a small testing budget. The separation family makes the gain visible with a simple comparison: static tests inspect B people one at a time, while a count-guided policy searches many more and certifies a healthy person when one is present. Structured posterior algorithms show that this information can be used exactly when overlap has small separators. The experiments then connect the two ends: exact counts improve the dynamic optimum, and a coarse three-level channel can capture most of that gain on a selected small instance.

A Monotonicity under Resolution Refinement

A quantizer Q partitions the possible counts. We restrict attention to quantizers that isolate count zero, because otherwise an observation cannot certify an all-clear pool under strict hard clearing.

Lemma 2 (Refinement never lowers optimal welfare). *If Q' refines Q, then $\mathcal{U}_Q \le \mathcal{U}_{Q'}$.*

Proof. Run an optimal Q-policy under Q'. At every stage, deterministically coarsen the Q' observation to its Q bin and choose the pool prescribed by the Q-policy. This reproduces the same coarsened history and the same tested pools on every latent profile. Since both channels isolate zero, the realized hard-clearing reward is identical. The optimal Q'-policy can only do better. $\square$

The argument is a sequential deterministic-garbling construction, related to the comparison of experiments [3]. With $B = 1$, no observation can affect a later action, so every zero-isolating resolution has the same optimal value. The recorded resolution data include this flat-horizon control.

B Posterior Complexity, Fibers, and Sampling

An exact-count history imposes linear constraints $Az = r$ on a binary latent profile z. The posterior normalizer is a weighted sum over the feasible fiber

$$\mathcal{F}(A, r) = \{z \in \{0,1\}^N : Az = r\}.$$

This representation explains both the informational value and the computational cost of count feedback. We do not establish a general complexity theorem here: both the complexity of the weighted normalizer and that of an individual posterior marginal remain open in this model.

A small enumeration remains useful for intuition. For pools $\{0, 1, 2\}$ and $\{2, 3, 4\}$, both observed with exact count one, only five profiles satisfy the two equations. Replacing the exact outcomes with two positive binary outcomes leaves 25 profiles. The all-infected profile belongs to the binary set but not to the exact-count fiber. The comparison isolates how exact counts sharpen the posterior, complementing the welfare separation in the main text.

The implementation uses exact enumeration inside small connected components and a Metropolis-style fallback for larger components. Local swap moves alone are not sufficient: they preserve total infection count and can leave a fiber disconnected. Alternating-path proposals repair verified small counterexamples, but the prototype sampler does not prove irreducibility on every binary fiber. A general sampler claim requires a connecting move set or a scoped connectivity theorem.

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